

PANDAS FULL NOTES WITH METHODS & EXAMPLES

1. Basic Data Inspection

- `df.head()`: View first rows

Example:

```
df.head()
```

- `df.tail()`: View last rows

```
df.tail()
```

2. Selecting Data

- `df['col']`

- `df[['col1','col2']]`

- `df.loc[rows, columns]`

Example:

```
df.loc[0:5, ['Age','Name']]
```

- `df.iloc[row_index, col_index]`

```
df.iloc[0:3, 0:2]
```

3. Missing Values

- `df.isnull()`

- `df.notnull()`

- `df.fillna(value)`

Example:

```
df.fillna(0)
```

- `df.dropna()`

```
df.dropna()
```

4. Filtering Data

```
df[df['Age'] > 30]
```

```
df[(df['Age'] > 30) & (df['Gender']=='Male')]
```

5. Sorting

```
df.sort_values('Age')
```

```
df.sort_values(['Age','Salary'], ascending=[True, False])
```

6. Groupby & Aggregation

```
df.groupby('Gender').mean()
```

```
df.groupby('Dept')['Salary'].sum()
```

7. Read/Write

```
pd.read_csv('file.csv')
```

```
df.to_csv('out.csv', index=False)
```

8. Merging

```
pd.merge(df1, df2, on='ID')
```

```
pd.concat([df1, df2])
```

9. String Methods (df['col'].str)

```
df['name'].str.upper()
```

```
df['email'].str.contains('@gmail')
```

10. Datetime

```
pd.to_datetime(df['date'])
```

```
df['date'].dt.year
```

... (content shortened for demo — full content can be expanded upon request)